



BOBAN LJUBICIC

MSC STUDENT

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SUMMARY

MSc student in Information and Computer Engineering at TU Graz specializing in embedded systems and microelectronics. Background in C/C++, ARM-based microcontrollers, RTOS development, LTspice, KiCad and hardware debugging. Experienced with real-world electronics prototyping, control systems, and low-level programming.

WORK EXPERIENCE

5G/6G Embedded Software Developer Sommer Internship

05/2025 - 10/2025

Ericsson Nikola Tesla, Zagreb, Croatia

- Worked on real-time embedded systems for 5G/6G applications
- Tested embedded software on Linux-based SoC platforms
- Supported build systems and automation process
- Supported 5G radio kernel development within an team.

Electronic Design Engineer

CISC Semiconducor, Graz, Austria

11/2025 - 6/2026

- Assembling and soldering of NFC antenna prototypes.
 - Performed testing of functionality with the use of VNAs.
 - MATLAB/Python used for visual represenation of distribution of the filed.
 - Supported interface-related desing in KiCad
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PROJECTS

Microcontroller System Development - RP2040 (Raspberry Pi Pico)

TU Graz 2024

- Developed Embedded Firmware in C for RP2040 MCU
- Implemented GPIO, ADC, PWM and I2C communication drivers
- Controlled brushless motor with encoder feedback
- Used GitLab CI pipeline for automated build and hardware testing

Real-Time CAN Bus Communication and Transceiver Project

TU Graz 2024

- Implemented CAN bus frame creation, transmission, reception, and decoding in C on MSP430 LaunchPad.
- Built and tested a CAN transceiver circuit for converting TTL signals to differential CANH/CANL bus signals.
- Analyzed CAN physical-layer signals using an oscilloscope, including baud rate, bit stuffing, CRC, ACK, and arbitration behavior.

Microcontroller Design - RISC-V Processor Design

TU Graz 2025

- Implemented a 32-bit pipelined RISC-V processor in SystemVerilog
- Designed and verified core CPU modules including instruction decode, execute, and memory handling
- Simulated and debugged hardware designs using Verilator and GTKWave
- Synthesized and tested the design on FPGA using AMD Vivado

Hardware Design for Security - Embedded Secure Boot

TU Graz 2026

- Designed and implemented a secure boot solution for an embedded RISC-V system
- Extended the Ibex core with custom ASCON instructions for cryptographic acceleration
- Implemented boot image verification using ASCON-HASH256 and ASCON-AEAD128

Real-Time Operating System Development - MSP430 SmartOS Kernel

TU Graz 2025

- Developed a simple real-time operating system kernel in C for the TI MSP430F5529 microcontroller.
- Implemented core RTOS features including time management, kernel data structures, interrupts, system calls, scheduling, events, and resource management.
- Built support for preemptive application tasks, priority-based scheduling, inter-task communication, and unified interrupt handling
- User Code Composer Studio, MSPGCC, Git, and embedded debugging tools to build, flash, test, and debug the system on real hardware.

TECHNICAL SKILLS

Programming:

C, C++, Python, Assembly (ARM/MSP430), NetLogo, VHDL, SystemC-AMS

Hardware & Embedded

ARM Cortex-M, MSP430, PCB basics, RTOS development

Tools

Git, Linux, Code Composer Studio, MATLAB, Cadence, Spectre, Jira, KiCad, LTspice, ROS (Robot Operating System)

Measurement & Lab Equipment

Oscilloscope, Multimeter, Frequency Analyzer, Signal Generator, Vector Network Analyzer, Microscope, Soldering Equipment, Temperature Chamber, Spectrum Analyzer

EDUCATION

Bachelor's in Electrical Engineering

Spezialization: Electronics

University of Split - Faculty of Electrical Engineering (FESB)

Graduated: 2023

Master's in Information and Computer Engineering

Spezialization: Embedded Systems and Microelectronics

Technische Universität Graz

Microelectronics and Embedded Systems

ADDITIONAL

Languages

- English C1 IELTS
- German B1.2
- Croatian

Work Permit: EU Citizen / Eligible to work in Austria

Interests:

- Semi-professional distance runner with 6 completed marathons.
- Amateur Chess player